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```
import numpy as np
import pandas as pd

df=pd.read_csv('Iris.csv')
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   sepal.length    150 non-null    float64
 1   sepal.width     150 non-null    float64
 2   petal.length    150 non-null    float64
 3   petal.width     150 non-null    float64
 4   variety         150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB

df.variety.value_counts()

variety
Setosa      50
Versicolor  50
Virginica   50
Name: count, dtype: int64

df.head()

   sepal.length  sepal.width  petal.length  petal.width  variety
0           5.1           3.5           1.4           0.2   Setosa
1           4.9           3.0           1.4           0.2   Setosa
2           4.7           3.2           1.3           0.2   Setosa
3           4.6           3.1           1.5           0.2   Setosa
4           5.0           3.6           1.4           0.2   Setosa

features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2
,random_state=42)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

KNeighborsClassifier()
```

```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

0.9666666666666667
1.0

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  1, 49]])
```